

LLM with Construction Engineering Prior Knowledge using agentic graphRAG

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Abstract—On large construction projects, engineering instructions and method statements must travel through a long chain of members—from the client to the site engineer, the foreman, and finally the workers on site. In practice this chain is lossy: information is fragmented across chat applications and paper records, critical numeric tolerances in repair procedures are misread, and Thai–English language barriers with a multinational workforce introduce further ambiguity, all of which translates into rework and non-conformance reports. We present SmartCivil, a question-answering framework that couples a large language model (LLM) with a knowledge graph to give site members reliable, source-grounded access to engineering knowledge. Scanned bilingual method statements are ingested by a vision-language OCR model; the extracted text, headings, tables, and figures populate a knowledge graph whose entities and relations make the engineering knowledge explicit and queryable. For each user question the system performs hybrid retrieval—keyword, full-text, and dense semantic search—whose rankings are merged by reciprocal rank fusion, and an LLM synthesises an answer that is constrained to the retrieved context and annotated with page-level citations. We evaluate the framework on a tunnel-segment repair corpus using both single-category and multi-category questions, scoring engineering-answer correctness, document grounding, bilingual fidelity, and response latency. The results characterise where retrieval-augmented generation succeeds and where document grounding remains the limiting factor for safety-critical construction question answering.

Index Terms—large language models, retrieval-augmented generation, knowledge graph, optical character recognition, hybrid retrieval, reciprocal rank fusion, bilingual question answering, construction informatics

I. INTRODUCTION

Large language models (LLMs) have become strong general-purpose reasoners over natural language, yet when deployed as domain question-answering systems they exhibit a well-known failure mode: they hallucinate [3], producing fluent answers that are not supported by any authoritative source. In safety-critical engineering domains, where a single misquoted tolerance can invalidate a repair procedure, ungrounded generation is unacceptable. Retrieval-augmented generation (RAG) [2] mitigates this by conditioning the model on passages retrieved from a trusted corpus, so that generation is anchored to verifiable evidence rather than parametric memory.

The effectiveness of RAG is bounded by the quality of its retrieval stage. Sparse lexical retrieval such as BM25 [4] rewards

exact term overlap and is robust for rare technical identifiers (for example, repair-method codes), whereas dense retrieval over learned embeddings [5] captures semantic similarity and paraphrase. Because the two signals are complementary, we adopt a hybrid retriever and combine its component rankings with reciprocal rank fusion (RRF) [6], a rank-based aggregation that requires no score calibration across retrievers. Beyond flat passage retrieval, a knowledge graph [7] imposes explicit structure on the corpus—entities such as defects, materials, and procedures, together with the relations among them—providing a grounding substrate that supports multi-hop and constraint-bearing queries that plain chunk retrieval handles poorly.

Two further technical challenges arise from the source material. First, the engineering knowledge exists only as scanned, image-based method statements, so the corpus must be reconstructed with optical character recognition (OCR) using a vision-language model [8] that can recover text, headings, and tabular structure. Second, the documents and the user population are bilingual (Thai and English) and frequently code-switched, which requires retrieval and generation to preserve technical terminology across languages. We further employ query decomposition for compound questions and constrain the generator to emit structured, citation-bearing answers so that every claim is traceable to a source page.

This paper makes the following technical contributions:

- a bilingual document-ingestion pipeline that converts scanned method statements into a structured, retrievable knowledge base via vision-language OCR;
- a hybrid keyword, full-text, and dense retriever whose rankings are fused by RRF and consumed by an LLM under a grounding constraint with page-level citation; and
- a multi-dimensional evaluation protocol—covering engineering-answer correctness, document grounding, bilingual fidelity, and latency—applied to a tunnel-segment repair corpus.

II. METHODOLOGY

Building on the RAG-LLM design of POLRAG [1], SmartCivil is organised as a six-phase pipeline that transforms scanned, image-based method statements into grounded, citation-bearing answers. The phases are document ingestion,

knowledge-graph construction, query understanding, retrieval, answer generation, and structured formatting.

A. Document Ingestion

The source corpus consists of scanned bilingual (Thai–English) method statements for tunnel-segment repair. Each page is processed by a vision-language OCR model that recovers running text, headings, tables, and figures while preserving the page index. The output is serialised into a manifest together with extracted image and table assets, so that every text segment retains a back-pointer to its originating page; these page references are later surfaced as citations and are the basis of the document-grounding evaluation.

B. Knowledge-Graph Construction

[Incomplete — to be written.]

C. Query Understanding

[Incomplete — to be written.]

D. Retrieval

For each (sub-)question the system runs three retrievers in parallel. A keyword retriever scores each segment D against the query Q with Okapi BM25, which rewards shared terms while damping common words and long segments,

$$\text{BM25}(Q, D) = \sum_{q \in Q} \text{IDF}(q) \frac{tf(q, D)(k_1 + 1)}{tf(q, D) + k_1 \left(1 - b + b \frac{|D|}{\text{avgdl}}\right)}, \quad (1)$$

where $tf(q, D)$ is the term frequency, $|D|$ the segment length, avgdl the mean segment length, $\text{IDF}(q)$ the inverse document frequency, and $k_1 = 1.5$ and $b = 0.75$ control term-frequency saturation and length normalisation. A full-text retriever applies the same term-based scoring over both Thai and English surface forms. A dense retriever ranks segments by the cosine similarity between the question embedding $\mathbf{q}$ and each segment embedding $\mathbf{d}$,

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|}. \quad (2)$$

The three rankings are merged by reciprocal rank fusion (RRF), which aggregates on rank position and therefore needs no score normalisation across retrievers,

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)}, \quad (3)$$

where R is the set of retrievers, $\text{rank}_r(d)$ is the rank of segment d under retriever r , and k is a smoothing constant. The top-ranked fused segments form the context passed to the generator.

E. Answer Generation and Structured Formatting

A system prompt constrains the LLM to answer strictly from the retrieved context, to decline when the context is insufficient, and to respond in the language of the question. The raw answer is then passed through a formatting step that filters OCR noise and emits a structured response: the textual answer, any supporting image or table evidence, and the page-level document citation.

III. EXPERIMENT DESIGN

We evaluate two complementary aspects of the system: the quality of the retrieval stage in isolation, and the end-to-end quality of the generated answers, both measured on a synthetic question–answer set.

A. Synthetic QA Dataset

The evaluation set is generated synthetically from the source repair manual by prompting an LLM to act as a domain QA-pair generator. The generator is constrained to use *only* information explicitly stated in the document—no external knowledge or general engineering practice—to include exact numeric values, to attach the originating page number to every answer, and to abstain when the evidence is unclear. Questions and answers are produced in the language of the source (Thai or English, frequently code-switched).

Each item carries a two-tier label. A *Rainbow Group* marks whether the question is Construction-Related (Group A) or Non-Construction (Group B, e.g. document or project metadata). Construction-Related items are further assigned a category—Repair Method, Material, Human Risk, or Environmental/Preventive Measures—each mapped to specific manual sections (Table I).

TABLE I
RAINBOW-GROUP CATEGORY TAXONOMY

Category	Source manual sections
Repair Method	Guideline for Repair; Cementitious Mortar; Epoxy Resin; Spalling Edge; Voids and Blow-holes
Material	Material Data Sheet
Human Risk	Risk Assessment; Material Safety Data
Environmental / Preventive	Preventative Measures and Environmental Impact
Non-Construction	Document/project metadata (Group B)

Two difficulty levels are evaluated, with 20 questions each:

- **Level 1 (single-category):** each question targets exactly one category and is answerable from a single section.
- **Level 2 (multi-category):** each question deliberately combines two or three categories and is multi-part, requiring evidence assembled from several sections—for example, selecting the correct repair method, then its material mixing ratio, then the associated environmental measure.

B. Retrieval Evaluation

Because segment-level relevance labels are unavailable, retrieval is assessed at the page level against the ground-truth page(s) attached to each question. Let $\mathcal{R}$ be the set of distinct pages among the top- k retrieved segments and $\mathcal{G}$ the ground-truth page set; the metrics are

$$\text{Recall} = \frac{|\mathcal{R} \cap \mathcal{G}|}{|\mathcal{G}|}, \quad \text{Precision} = \frac{|\mathcal{R} \cap \mathcal{G}|}{|\mathcal{R}|}, \quad (4)$$

$$\text{Hit}@k = \mathbb{I}[\mathcal{R} \cap \mathcal{G} \neq \emptyset].$$

Recall measures how completely the retriever surfaces the evidence pages, $\text{Hit}@k$ whether at least one correct page

appears in the top- k , and precision is bounded below because the system returns k pages where typically only one is relevant. These page-level scores serve as a practical proxy for context recall and precision.

C. LLM Performance

End-to-end answer quality is scored on a 100-point rubric spanning five dimensions, summarised in Table II. Engineering-answer correctness and document-grounding accuracy carry the largest weights because they capture, respectively, whether the prescribed repair method matches the method statement and whether the answer is faithful to the source documents without hallucination.

TABLE II
LLM PERFORMANCE EVALUATION RUBRIC

Evaluation Dimension	Max	Metric / Method
Engineering Answer Correctness	40	Expert validation, rule matching
Document Grounding Accuracy	30	Citation correctness, hallucination rate
Bilingual Capability (TH-EN)	10	Translation accuracy, semantic consistency
Response Latency	10	Average response time (s)
System Prompt Compliance	10	Prompt adherence score
Total	100	

Each scored dimension is graded into qualitative bands. For engineering-answer correctness, an answer is *Excellent* (36–40) when it identifies the correct repair method in line with the method statement (e.g. RP-TN-01, RP-TN-02, or Reject) together with the complete numeric conditions such as crack width, depth, and void size; *Good* (26–35) when the method is correct but some numeric detail is missing; *Fair* (16–25) when the intent is partially correct but the chosen method departs from the guideline; and *Poor* (0–15) when the method is wrong or contradicts the documents. For document-grounding accuracy, an answer is *Excellent* (26–30) when it is entirely supported by the corpus with no hallucination, *Good* (18–25) when citations are largely correct with minor over-summarisation, and *Poor* (0–17) when it contains clear hallucination or unsupported claims. Response latency is mapped from the mean per-question time: ≤ 5 s scores 10, 6–10 s scores 8, 11–20 s scores 5, and > 20 s scores 0.

IV. RESULTS AND DISCUSSION

We evaluate on two question sets: a single-category set (Level 1, $n = 20$) and a multi-category set that pairs two or three topics per question (Level 2, $n = 20$). The reported configuration uses multilingual dense embeddings for the semantic retriever and disables LLM query rewriting, a setting selected because an ablation showed rewriting degraded retrieval (it lowered keyword Hit@ k on Level 2 from 0.75 to 0.55).

A. Retrieval

Table III reports the page-level recall, precision, and Hit@ k defined in (4) for each retriever. The dense (*vector*) retriever is the strongest single method, and RRF fusion gives the best result on Level 1; on Level 2 the vector retriever alone is best, as the weaker keyword ranking slightly drags the fusion down. Precision is uniformly low because the system returns $k = 5$ pages per query while most answers reside on a single page, so recall and Hit@ k are the informative signals. Crucially, dense retrieval is decisive only with genuine semantic embeddings: with a non-semantic baseline embedding the vector retriever collapsed (Level 2 Hit@ k 0.40) and fusion added nothing, whereas with real embeddings it reaches 0.95.

TABLE III
RETRIEVAL QUALITY BY METHOD (PAGE-LEVEL)

Retriever	Level 1			Level 2		
	Rec.	Prec.	Hit	Rec.	Prec.	Hit
Keyword (BM25)	0.45	0.10	0.45	0.57	0.17	0.75
Vector (dense)	0.65	0.15	0.65	0.77	0.23	0.95
Fused (RRF)	0.75	0.17	0.75	0.67	0.20	0.85

B. End-to-End Answer Quality

Each fused-retrieval answer was scored against an expert ground truth on the rubric of Table II; latency was derived from the measured per-question time, and bilingual capability and system-prompt compliance were assessed at the system level. Table IV reports the aggregate.

TABLE IV
EVALUATION RESULTS (FUSED RETRIEVAL)

Dimension	Max	L1	L2
Engineering Answer Correctness	40	33.0	32.9
Document Grounding Accuracy	30	23.1	23.4
Response Latency	10	3.2	0.8
Bilingual Capability (TH-EN)	10	9.0	9.0
System Prompt Compliance	10	9.0	9.0
Total	100	77.3	75.0

Engineering-answer correctness is high on both sets (33.0 and 32.9 of 40): when the right page is retrieved, the model identifies the correct repair method and quantitative conditions, and multi-category questions are usually answered across all their sub-parts. The residual gap comes from a few honest abstentions on questions whose evidence was not retrieved, and from occasional omission of a second numeric sub-part (e.g. a mixing ratio or pot life).

Table V contrasts a high- and a low-scoring answer (both translated from Thai). The Excellent answer reproduces the correct method- statement criteria with complete numeric bounds and a correct page citation; the Poor answer is fluent and well-formed but reports a numeric value that contradicts the data sheet, which is precisely the failure mode the correctness dimension is designed to penalise.

TABLE V
EXAMPLE ANSWERS AT HIGH AND LOW CORRECTNESS

Excellent — 38/40	
Question	Repair criteria for a “Hard Crack” repaired with Cementitious Mortar?
Ground truth	width $0.1 < A < 0.3$ mm and depth > 15 mm.
System	“width (A) > 0.1 mm but < 0.3 mm; depth > 15 mm” — cites <i>Guideline for Repair</i> , p. 3.
Why	correct criteria, complete numeric bounds, correct citation.
Poor — 10/40	
Question	Bond strength to concrete of Sikadur-32TH?
Ground truth	2.5–3 N/mm ² (concrete failure).
System	“ > 4 N/mm ² for dry and moist concrete.”
Why	wrong numeric value; contradicts the data sheet despite fluent form.

C. Document Grounding and Citation

Grounding scores in the upper *Good* band (23.1 and 23.4 of 30): answers are overwhelmingly faithful to the corpus and carry chapter/page citations, but the score is capped by noisy citations—the system returns several candidate pages where typically one is correct, and a handful of otherwise-correct answers cite an adjacent page. Hallucination is rare. Table VI illustrates the distinction from correctness: a *factually correct* answer can still score low on grounding when it cites the wrong source page.

TABLE VI
EXAMPLE ANSWERS AT HIGH AND LOW GROUNDING

Excellent — 27/30	
Question	First aid if Sika Skim Coat enters the eye?
Ground truth	remove contact lenses, rinse with water ≥ 10 min, see a doctor (p. 125).
System	states the same three steps and cites <i>Materials Safety Data</i> , p. 125.
Why	fully supported by the corpus, correct page citation, no hallucination.
Poor — 15/30	
Question	Project name stated in the document?
Ground truth	Sewerage Tunnel Bung Nong Bon to Chaophraya River (p. 1).
System	names the correct project but cites p. 147 (a material-approval page).
Why	fact is right, but the cited page is unsupported (page-hit miss); grounding penalises the citation error.

D. Latency

Latency remains the weakest dimension (mean 19.7 s on Level 1, 28.7 s on Level 2), because each query is answered independently for every retrieval mode and passes through a second answer-formatting stage. This is the most actionable deficit; restricting to the fused mode and dropping the second pass would reduce it substantially.

E. Bilingual Fidelity and Compliance

[Pending: bilingual evaluation needs Thai-language examples, which cannot be typeset in this pdfL^AT_EX build for now.] Bilingual capability and prompt compliance are strong: answers consistently match the language of the question, preserve English material and technical terms, cite their sources, and abstain when the context is insufficient. The defining compliance behaviour is correctly *declining* an unanswerable question rather than guessing; Table VII contrasts this with the opposite failure, where the system over-generates beyond the retrieved context.

TABLE VII
EXAMPLE ANSWERS AT HIGH AND LOW PROMPT COMPLIANCE

Excellent — 10/10	
Question	Pot life of Sikadur-31 CF Normal at +30°C? (not in the corpus)
System	“the provided document does not contain this information.”
Why	obeys “answer only from context, decline when insufficient”; stays in scope, correct language and format.
Poor — 3/10	
Question	Project name stated in the document? (earlier configuration)
System	returns a list of ~ 25 unrelated project names instead of the single relevant answer or a decline.
Why	out-of-scope over-generation; violates “answer strictly from the retrieved context.”

Overall the framework is reliable when retrieval succeeds; with semantic embeddings in place, latency and citation precision are the remaining priorities for future work.

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